

Release Notes

Version 1.0.0 - Initial Release

Release Date: 2023-10-01

Status: Stable

Summary

This release introduces the initial version of the expense management application for our organization. The application allows employees to submit expenses, managers to approve or reject them, and provides a reporting dashboard for viewing expenses. The application is built using a microservices architecture with a PostgreSQL database, Spring Boot for the backend, and a RESTful API for communication.

New Features

- **Expense Submission:** Employees can submit their expenses through the application, which are then stored in the database for further processing. This feature enables employees to easily track and submit their expenses without manual paperwork.
- **Expense Approval Workflow:** Managers can approve or reject submitted expenses, updating the status accordingly. This feature streamlines the approval process and ensures that expenses are reviewed and processed efficiently.
- **Reporting Dashboard:** Users can view a list of all expenses and retrieve specific expense details. This feature provides transparency and allows for better financial oversight.

Core Capabilities Delivered

- **Database Schema:** The application includes a PostgreSQL database schema designed to store employee and expense data efficiently.
- **RESTful API:** A set of RESTful APIs is provided for interacting with the expense management system, including endpoints for submitting, retrieving, updating, and deleting expenses.
- **Security:** The application implements JWT-based authentication and role-based access control to ensure secure access to the system.

Improvements & Polish

- **Input Validation:** Custom validation annotations are used to ensure that all input data meets the required criteria, reducing the risk of invalid data being processed.
- **Rate Limiting:** Rate limiting is implemented to prevent abuse of the API endpoints, ensuring the system remains responsive and available.
- **UI Refinement:** The user interface has been refined to provide a better user experience, making it easier for employees and managers to interact with the application.

Known Issues & Limitations

- **Third-Party Integrations:** Integration with third-party services is not yet implemented. This will be addressed in version 1.1.
- **Performance:** The application currently handles up to 1000 records efficiently, but performance may degrade with larger datasets. Optimizations will be implemented in future versions.

Breaking Changes

- **None:** This initial release does not introduce any breaking changes.

Compatibility

- **Requires:** Java 11, PostgreSQL 13, Spring Boot 2.5.x
- **Tested on:** Ubuntu 20.04, Windows 10, Google Chrome, Mozilla Firefox

Upgrade Notes

To deploy this release:

1. Back up the database
2. Pull the latest code: `git pull origin main`
3. Install/update dependencies: `mvn clean install`
4. Run migrations if any: `mvn liquibase:update`
5. Restart services
6. Verify health: `curl http://localhost:8080/api/health`

Contributors

- **John Doe:** Lead Developer, implemented core features and API endpoints
- **Jane Smith:** Database Administrator, designed and implemented the database schema
- **Alice Johnson:** Security Engineer, implemented authentication and authorization

What's Next

Planned for Version 1.1

- **Third-Party Integrations:** Implement integration with third-party services such as payroll and accounting systems.
- **Performance Improvements:** Optimize database queries and API endpoints to handle larger datasets.
- **Enhanced Reporting:** Add additional reporting features, such as expense summaries and trend analysis.

Future Roadmap

- **Mobile Application:** Develop a mobile application for expense submission and approval on-the-go.
- **Machine Learning:** Implement machine learning algorithms to detect and flag suspicious expenses.

Feedback & Support

- **Report issues:** Submit issues on GitHub or via email at support@example.com
- **Request features:** Submit feature requests on GitHub or via email at support@example.com
- **Documentation:** See README.md, User Guide, and API Documentation